Siyuan Zhou

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Education

Fudan University Shanghai, China
B.S. in Data Science (GPA: 3.95, Rank: 2/76) Sept. 2023 - Present
University of California, Berkeley (Exchange) Berkeley, CA
Statistics (GPA: 4.00) Oct. 2025 - Dec. 2025

Research Interests

Structured Representation Learning, Generative Modeling, Model Editing, Trustworthy & Explainable AI

Publications

- **Siyuan Zhou**, Zhibin Gu. "OPTION: Optimal Transport–Guided Flow Matching for Incomplete and Unaligned Multi-View Clustering." *Submitted to 43rd International Conference on Machine Learning **(ICML 2026)***.
- Yuanye Liu*, **Siyuan Zhou***, Ke Zhang, Lei Li, Wei Chen, Xiahai Zhuang. "X-Edit: Exact, Explicit, and Explainable Null-Space Editing for Medical Vision Transformers." *Submitted to International Conference on Medical Image Computing and Computer Assisted Intervention **(MICCAI 2026)***.

Research Experience

Gu Lab, Hebei Normal University Shijiazhuang, China
Research Assistant *(Mentor: Prof. Zhibin Gu)* Oct. 2025 - Present

Optimal Transport–Guided Flow Matching for Incomplete and Unaligned Multi-View Clustering

This project develops a unified framework for multi-view clustering under realistic settings with missing views and cross-view misalignment. We formulate generation, alignment, and clustering as a single end-to-end optimization problem, leveraging conditional flow matching for efficient missing-view imputation and Gromov–Wasserstein optimal transport for alignment-free structural fusion. The resulting system learns geometry-preserving latent representations and achieves robust clustering performance without requiring explicit cross-view correspondences.

- Conducted a comprehensive study of multi-view learning, optimal transport, and generative modeling, and identified key limitations in existing two-stage pipelines for incomplete and unaligned settings.
- Designed a unified architecture integrating latent representation learning, conditional flow matching (ODE-based generation), and Gromov–Wasserstein structural alignment within a single objective.
- Implemented an efficient flow-matching-based imputation module, achieving deterministic generation with significantly reduced inference cost compared to diffusion-based approaches.
- Developed scalable batch-wise structural alignment via kernel-based GW loss, enabling correspondence-free multi-view fusion.
- Conducted extensive experiments on multiple benchmark datasets, demonstrating strong robustness under high missingness and severe misalignment.
- Outcome: ICML 2026 submission.

Scalable Structure-Preserving Multi-View Clustering with Optimal Transport and Density Estimation

This project develops a scalable framework for multi-view clustering under heterogeneous feature spaces and varying view quality. We focus on preserving intrinsic geometric structure while achieving near-linear computational complexity. The framework integrates sparse feature learning, density-aware view weighting, and optimal-transport-based graph refinement to enable robust and efficient clustering on large-scale multi-view data.

- Conducted a systematic study of multi-view learning, kernel density estimation, and optimal transport, identifying scalability bottlenecks in existing clustering pipelines.
- Designed a structure-preserving sparse feature selection module using randomized SVD and Laplacian regularization to efficiently learn low-dimensional representations.
- Developed a density-aware view weighting mechanism based on k-NN kernel density estimation, enabling adaptive handling of heterogeneous and noisy views.
- Proposed a landmark-based optimal transport formulation to refine graph structure with near-linear complexity, avoiding quadratic transport costs.
- Integrated local (k-NN graph) and global (OT-based) geometric information via graph enhancement, improving clustering robustness and consistency.
- Implemented a fully scalable pipeline with near-linear time and memory complexity, and validated its performance on large-scale multi-view datasets.

ZMIC Lab, Fudan University Shanghai, China
Research Assistant *(Mentor: Prof. Xiahai Zhuang)* Dec. 2025 - Present
Exact and Explainable Null-Space Model Editing for Medical Vision Transformers
This project develops a theoretically grounded model editing framework for correcting failure cases in medical Vision Transformers without catastrophic forgetting. We formulate post-hoc model intervention as a constrained optimization problem, combining causal tracing for precise localization of error-relevant components with null-space projections that guarantee preservation of previously learned representations. The resulting framework enables exact, efficient, and interpretable parameter updates with strong reliability for clinical deployment.

- Conducted a systematic study of model editing, continual learning, and interpretability in deep neural networks, identifying the limitations of gradient-based fine-tuning in safety-critical settings.
- Designed a null-space constrained editing formulation, transforming parameter updates into a closed-form solution that guarantees no interference with preserved knowledge.
- Developed a causal tracing procedure to localize influential layers responsible for erroneous predictions, enabling targeted and interpretable interventions.
- Implemented a closed-form update mechanism for efficient post-hoc editing, eliminating the need for iterative optimization and significantly reducing computational cost.
- Extended the framework to support sequential editing while maintaining stability across previously edited samples.
- Conducted extensive experiments on multiple medical imaging benchmarks, demonstrating near-zero performance degradation, high edit success rates, and superior efficiency compared to existing methods.
- Outcome: MICCAI 2026 submission.

Zhu Lab, Fudan University Shanghai, China
Research Assistant *(Mentor: Prof. Shanfeng Zhu)* Apr. 2024 - Apr. 2026
Deep Learning–Based Viral Sequence Identification and Classification Pipeline
This project investigates computational methods for viral sequence identification and classification in metagenomic data, aiming to bridge the gap between rapidly growing sequencing data and limited annotation capacity. We systematically analyze existing tools and develop a unified pipeline that integrates multiple algorithms to improve classification accuracy, robustness, and standardization.

- Conducted a comprehensive study of state-of-the-art viral identification and classification tools, including deep learning–based (e.g., DNABERT-based models), alignment-based, and graph-based methods.
- Evaluated representative tools (e.g., ViraLM, CheckV, 4CAC, Genomad, Minimap2, PhaCGN) across multiple datasets, comparing their performance on accuracy, robustness, and scalability.

- Analyzed key challenges in viral sequence analysis, including short contig classification, dataset bias, inconsistent taxonomy standards, and tool-specific limitations.
- Identified trade-offs between methods: alignment-based approaches provide higher accuracy and stability, while deep learning models offer better generalization but struggle with fine-grained classification.
- Designed a unified analysis pipeline integrating multiple tools and standardizing outputs, with plans for weighted ensemble strategies to improve overall classification performance.

Honors & Awards

- National Scholarship 2024, 2025
- Second Prize, National Undergraduate Mathematical Contest in Modeling (Provincial Level) 2024
- Third Prize, National Undergraduate Mathematics Competition 2023, 2024
- First Prize, Fudan She Cup Academic English Vocabulary Competition (Undergraduate & Graduate Division) 2024, 2025
- Bronze Medal, China International College Students' Innovation Competition 2025

Technical Skills

- **Programming Languages**: Python, Shell/Bash scripting, C/C++, and HTML/CSS.
- **Machine Learning**: PyTorch, NumPy, SciPy, scikit-learn, and Pandas.
- **System Administration & DevOps**: Linux/cloud server administration, WSL2-based development, Docker, network configuration (e.g., NAS setup), and web deployment.
- **Research & Documentation**: LaTeX and Markdown for academic writing, version control with Git.
- **Visualization & AI Tools**: Framework Visualization, Diagram Design (PowerPoint, Canva), Prompt Engineering, Large Language Model (LLM) Utilization (ChatGPT, Gemini), AI-assisted Research and Development

Language Proficiency

- **English**: TOEFL 101 (Listening 25, Reading 27, Writing 26, Speaking 23)